

7. SEQUENTIAL OUTPUT TRACING

In this type of questions, a message comprising of randomised words or numbers is given as the input followed by steps of rearrangement to give sequential outputs. The candidate is required to trace out the pattern in the given rearrangement and then determine the desired output step, according as is asked in the questions.

Example : **Study the following information to answer the given questions :**

A word arrangement machine, when given an input line of words, rearranges them following a particular rule in each step. The following is an illustration of input and the steps of rearrangement : (S.B.I.P.O. 1095)

Input : As if it on an Zoo figure Of in at

Step I : an As if it on Zoo figure Of in at

Step II 2 an As at if it on Zoo figure Of in

Step III : an As at figure if it on Zoo Of in

Step IV : an As at figure if in it on Zoo Of

Step V : an As at figure if in it Of on Zoo

(and Step V is the last step for this Input).

As per the rules followed in the above steps, find out in the given questions the appropriate step for the given input.

1. Which of the following will be Step II for the given input ?

Input : am ace all if Is

- (a) ace all am Is if (ft) all am ace if Is (c) Is if am ace all
(rf) ace all am if Is (e) None of these

2. Input : you are at fault on this

Which of the following steps would be — are at fault on you this ?

- (a) I (6) II (c) III (d) IV (e) V

3. Input : Him and His either or her

Which ^Vstep will be the last step for this input ?

- (a) I (6) II (c) III (d) IV (e) V

4. Step IV was like this — an apple at cot was red on one side

Which of the following will definitely be the input ?

- (a) was cot red an on at one apple side
(ft) cot an at apple was red on one side
(c)- apple at an cot was red on one side
(d) Cannot be determined
(e) None of these

Solution :

Clearly, in the given arrangement, the words have been arranged alphabetically in a sequence, altering the position of only one word in each step.

1. Clearly, we have :

Input: am ace all if Is

Step I : ace am all if Is

Step II: ace all am if Is

So, the answer is (rf).

2. **Input:** you are at fault on this

Step I: are you at fault on this

Step II: are at you fault on this

Step III: are at fault you on this

Step IV : are at fault on you this

So, the answer is (d).

3. **Input:** Him and His either or her

Step I : and Him His either or her

Step II: and either Him His or her

Step III : and either her Him His or

Since all the words in the given input have been arranged alphabetically upto Step III, so it is the last step

Hence, the answer is (c).

4. Tracing the output steps for each of the given inputs, we find that Step IV for (a) is the same as that given in the questions, while in (b) and (c), the desired output occurred at Step III.

Input: was cot red an on at one apple side

Step I s an was cot red on at one apple side

Step II ' an apple was cot red on at one side

Step III : an apple at was cot red on one side

Step IV : an apple at cot was red on one side

So, the answer is (a).

EXERCISE 7

Directions (**Questions 1 to 5**) : **Study the following information and answer the questions given below it:** (Bank P.O. 1995)

An electronic device when fed with the numbers, rearranges them in a particular order following certain rules. The following is a step-by-step process of rearrangement for the given input of numbers.

Input:	85	16	36	04	19	97	63	09
Step I:	97	85	16	36	04	19	63	09
Step II :	97	85	63	16	36	04	19	09
Step III :	97	85	63	36	16	04	19	09
Step IV:	97	85	63	36	19	16	04	09
Step V:	97	85	63	36	19	16	09	04

(For the given input step V is the last step).

1. Which of the following will be Step V for the given input ?

Input : 25 08 35 11 88 67 23

(a) 88 67 35 25 23 11 08

(b) 88 67 35 25 08 11 23

(c) 08 11 23 25 35 67 88

(d) 88 67 35 25 23 08 11

(e) None of these

2. Which of the following will be Step III for the given input ?

Input : 09 25 16 30 32 19 17 06

- (a) 32 09 25 16 30 19 17 06 (6) 32 30 09 25 16 19 17 06
(c) 32 30 25 09 16 19 17 06 (d) 32 25 09 16 30 19 17 06
(e) None of these

3. Which of the following will be the last step for the given input ?

Input : 16 09 25 27 06 05

- (a) I (6) II (c) III (d) TV (e) None of these

4. Which of the following will be the last step for the given input ?

Input : 03 31 43 22 11 09

- (a) IV (6) V (c) VI
(d) Cannot be determined (e) None of these

5. If the Step IV is as given below, which of the following was the input ?

Step IV : 92 86 71 69 15 19 06 63 58

- (a) 86 92 69 71 15 19 06 63 58 (6) 15 86 19 92 06 69 63 58 71
(c) 15 19 06 63 58 86 92 69 71 (d) Cannot be determined
(e) None of these

Directions (Questions 6 to 10) : Study the following information and answer the given questions :

(S.B.I.P.O. 1997)

A word arrangement machine, when given an input line of words, rearranges them following a particular rule in each step. The following is an illustration of input and the steps of rearrangement.

Input: Go for to Though By easy To Access at

Step I: Access Go for to Though By easy To at

Step II: Access at Go for to Though By easy To

Step III : Access at By Go for to Though easy To

Step IV : Access at By easy Go for to Though To

Step V : Access at By easy for Go to Though To

Step VI: Access at By easy for Go Though to To

Step VII : Access at By easy for Go Though To to

(and Step VII is the last step for this input)

As per the rules followed in the above steps, find out in the given questions the appropriate step for the given input.

6. Input : story For around on was He at

Which of the following will be Step IV for the given input ?

- (a) around at For He on was story (6) around at For He on story was
(c) around at For He story on was (d) around at He For story on was
(e) None of these

7. Input : every and peer to an for

Which of the following steps would be 'an and every for peer to' ?

- (a) II (6) III (c) IV (d) V (e) None of these

8. Input : Together over series on feast the so

Which of the following steps will be the last but one ?

- (a) II (6) III (c) IV (d) V (e) None of these

9. Input s Over Go For through at one
Which step number will be the last step of the above input ?
(a) III (6) V (c) VI (cf) VII (e) None of these
10. The Step II of an input is as follows :
and Do pet to on that
Which of the following would definitely be the input ?
(a) Do on pet to and that (ib) Do pet to and that on
(c) Do and pet to on that (d) Cannot be determined
(e) None of these

Directions (Questions 11 to 15) : Study the following information and answer the questions given below it : (Bank P.O. 1998)

The admission ticket for an exhibition bears a password which is changed after every clock hour based on set of words chosen for each day. The following is an illustration of the code and steps of rearrangement for subsequent clock hours. The time is 9 a.m. to 3 p.m.

Day's first password : First Batch — 9 a.m. to 10 a.m.

is not ready cloth simple harmony burning

Second Batch — 10 a.m. to 11 a.m.

ready not is cloth burning harmony simple

Third Batch — 11 a.m. to 12 noon

cloth is not ready simple harmony burning

Fourth Batch — 12 noon to 1 p.m.

not is cloth ready burning harmony simple

Fifth Batch — 1 p.m. to 2 p.m.

ready cloth is not simple harmony burning

and so on.

11. If the password for the first batch was — "rate go long top we let have", which batch will have the password — ^Mgo rate top long have let we"?
(a) Second (6) Third (c) Fourth (d) Fifth (e) None of these
12. Day's *first password* — "camel road no toy say me not".
What will be the password for fourth batch *i.e.* 12 noon to 1 p.m. ?
(a) road camel toy no not me say (6) no road camel toy not me say
(c) toy no road camel not me say (d) toy camel road no say me not
(e) None of these
13. If the batch 2 of the day has the password — "came along net or else key lot", what would be the password for batch 4 (*i.e.* 12 noon to 1 p.m.) ?
(a) net or came along else key lot (6) came or net along lot key else
(c) or net along came lot key else (d) along net or came else key lot
(e) None of these
14. If the password for 11 a.m. to 12 noon was — "soap shy miss pen yet the she", what was the password for the first batch ?
(o) pen miss shy soap she the yet (6) she miss pen soap yet the she
(c) soap pen miss shy she the yet (d) miss shy soap pen she the yet
(e) None of these

15. If the password for 6th batch *Le.* 2 p.m. to 3 p.m. is — "are trap cut he but say lap", what will be the password for 2nd batch *Le.* 10-11 a.m. ?
- (a) trap are he cut lap say but (6) he cut trap are lap say but
 (c) cut he are trap but say lap (cf) are he cut trap lap say but
 (e) None of these

ANSWERS

Questions 1 to 5

Clearly, in the given arrangement, the numbers have been arranged in descending order in a sequence, altering the position of only one number in each step.

1. (a): Input: 25 08 35 11 88 67 23
 Step I t 88 25 08 35 11 67 23
 Step II: 88 67 25 08 35 11 23
 Step III: 88 67 35 25 08 11 23
 Step IV: 88 67 35 25 23 08 11
 Step V: 88 67 35 25 23 11 08
2. (c): Input: 09 25 16 30 32 19 17 06
 Step I: 32 09 25 16 30 19 17 06
 Step II: 32 30 09 25 16 19 17 06
 Step III: 32 30 25 09 16 19 17 06
3. (6): Input: 16 09 25 27 06 05
 Step I : 27 16 09 25 06 05
 Step II: 27 25 16 09 06 05 •
- Since all the numbers in the given input have been arranged in descending order upto Step II, so it is the last step.
- 4.(6): Input: 03 31 43 22 11 09
 Step 11 43 03 31 22 11 09
 Step II: 43 31 03 22 11 09
 Step n i: 43 31 22 03 11 09
 Step IV s 43 31 22 11 03 09
 Step V: 43 31 22 11 09 03
5. (c): Input: 15 19 06 63 58 86 92 69 71
 Step I: 92 15 19 06 63 58 86 69 71
 Step II x 92 86 15 19 06 63 58 69 71
 Step III: 92 86 71 15 19 06 63 58 69
 Step IV: 92 86 71 69 15 19 06 63 58

Questions 6 to 10

In the given arrangement, the words have been arranged alphabetically in a sequence, altering the position of only one word in each step.

6. (c): Input: story For around on was He at
 Step 11 around story For on was He at
 Step II i around at story For on was He
 Step III: around at For story on was He
 Step IV: around at For He story on was
7. (6): Input: every and peer to an for
 Step I: an every and peer to for
 Step II i an and every peer to for
 Step III: an and every for peer to

8. (d): **Input:** Together over series on feast the so

Step I: feast Together over series on the so

Step II: feast on Together over series the so

Step III: feast on over Together aeries the bo

Step IV: feast on over series Together the so

Step V: feast on over series so Together the

Step VI: feast on over series so the Together

Clearly, Step VI is the last step and V is the last but one (second last).

9. (e): **Input:** Over Go For through at one

Step I: at Ovei* Go For through one

Step II: at For Over Go through one

Step III: at For Go Over through one

Step IV: at For Go one Over through

Since all the words in the given input have been arranged alphabetically uptil Step IV, so it is the last step.

10. (e): Clearly, none of the given inputs gives the desired output at Step II on rearrangement.

Questions 11 to 15

Clearly, the given arrangement is as under :

In the first step, the first three and the last three letters are written in a reverse order.

Then, the first four and the last three letters are written in a reverse order.

The process is repeated to obtain successive output steps.

11. (c): *First batch* : rate go long top we let have

Second batch : long go rate top have let we

Third batch : top rate go long we let have

Fourth batch : go rate top long have let we

12. (a): *First batch* : camel road no toy say roe not

Second batch : no road camel toy not me say

Third batch : toy camel road no say me not

Fourth batch : road camel toy no not me say

13. (d): *Second batch* : came along net or else key lot

Third batch ; or net along came lot key else

Fourth batch : along net or came else key lot

14. (6): *Third batch* : soap shy miss pen yet the she

Second batch : pen miss shy soap she the yet

First batch : shy miss pen soap yet the she

15. (c): *Sixth batch* : are trap cut he but say lap

Fifth batch : cut trap arc he lap say but

Fourth batch : he are trap cut but say lap

Third batch : trap are he cut lap say but

Second batch : cut he are trap but say lap